
ARTIFICIAL LIGHT AT NIGHT AND BAT ACTIVITY

ALONG A DISTANCE GRADIENT

Grand Teton National Park, 2022

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OVERVIEW

Study question: does light colour (red vs. white) affect bat acoustic activity, and does the answer depend on intensity or distance from the source?

Design: 14 acoustic monitoring stations along a distance gradient (<0.52 km to >2.1 km from source), red vs. white light crossed at five intensities (10, 30, 50, 70, 100%). Survey season: 2022. Eight focal species: Epfu, Laci, Lano, Myev, Mylu, Myvo, Myyu, Myci.

PRIMARY MODEL

Family: nbinom2 (log)
Formula: detections ~ color * intensity + mean_phase +
pct_nonforest + brightness_dark + jd_c +
I(jd_c^2) + dist_cat + (1 | site)
Zero infl: ~1
Data: data_env

AIC	BIC	logLik	-2*log(L)	df.resid
24276.0	24405.1	-12117.0	24234.0	3436

Random effects:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	0.301	0.5486

Number of obs: 3457, groups: site, 14

Dispersion parameter for nbinom2 family (): 0.603

Conditional model:

	Estimate	Std.Err	z value	Pr(> z)	
(Intercept)	1.2861400	0.6075	2.117	0.034264	*
colorW	0.0107081	0.0923	0.116	0.907680	
intensity30	0.5958819	0.1365	4.365	1.27e-05	***
intensity50	0.3537144	0.1427	2.480	0.013156	*
intensity70	0.3537966	0.0933	3.794	0.000148	***
intensity100	-0.0002213	0.1159	-0.002	0.998477	
mean_phase	0.2150853	0.1159	1.856	0.063473	.
pct_nonforest	2.1297329	1.1333	1.879	0.060206	.
brightness_dark	-0.0262685	0.0339	-0.776	0.437882	
jd_c	-0.0071307	0.0017	-4.254	2.10e-05	***
I(jd_c^2)	-0.0002928	0.0002	-1.704	0.088464	.
dist_catFar	0.9896498	0.4815	2.055	0.039849	*
dist_catFurther	1.2080207	0.5988	2.017	0.043648	*
dist_catMedium	1.1962086	0.6759	1.770	0.076774	.
colorW:intensity30	-0.7281190	0.1710	-4.258	2.06e-05	***
colorW:intensity50	-0.4971992	0.1455	-3.417	0.000634	***
colorW:intensity70	-0.1228107	0.1363	-0.901	0.367657	

```
colorW:intensity100 0.0893243 0.1339 0.667 0.504684
```

```
---
```

```
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Zero-inflation model:
```

```
      Estimate Std.Err z value Pr(>|z|)
(Intercept)  -21.66  858.41  -0.025    0.98
```

```
Reference levels: color = R (red), intensity = 10%, dist_cat = Close.
```

POST-HOC: COLOUR X INTENSITY

Estimated marginal means on the response scale (detections/night), averaged over distance. Holm-corrected pairwise contrasts.

```
=== color | intensity (response scale) ===
```

```
intensity = 10:
```

color	response	SE	df	asympt.LCL	asympt.UCL
R	13.1	2.3	Inf	9.4	18.4
W	13.3	2.3	Inf	9.5	18.6

```
intensity = 30:
```

color	response	SE	df	asympt.LCL	asympt.UCL
R	23.8	4.5	Inf	16.3	34.7
W	11.6	2.1	Inf	8.1	16.7

```
intensity = 50:
```

color	response	SE	df	asympt.LCL	asympt.UCL
R	18.7	3.3	Inf	12.5	27.8
W	11.5	2.0	Inf	8.0	16.5

```
intensity = 70:
```

color	response	SE	df	asympt.LCL	asympt.UCL
R	18.7	3.0	Inf	13.3	26.2
W	16.7	2.7	Inf	11.9	23.4

```
intensity = 100:
```

color	response	SE	df	asympt.LCL	asympt.UCL
-------	----------	----	----	------------	------------

R	13.1	2.3	Inf	9.2	18.7
W	14.5	2.4	Inf	10.3	20.4

=== Red vs. White contrast at each intensity (Holm-corrected) ===

intensity	ratio	SE	df	null	z.ratio	p.value	
10	0.985	0.129	Inf	1	-0.116	0.9077	
30	2.052	0.350	Inf	1	4.258	<.0001	***
50	1.630	0.237	Inf	1	3.417	0.0019	**
70	1.120	0.153	Inf	1	0.833	0.6112	
100	0.914	0.122	Inf	1	-0.667	0.6112	

Ratio = R / W (ratio > 1 = more detections under red)

At 30% intensity, red light produced 2.05x more detections than white ($p < 0.0001$). At 50%, 1.63x ($p = 0.002$). At 70% and 100%, no significant difference.

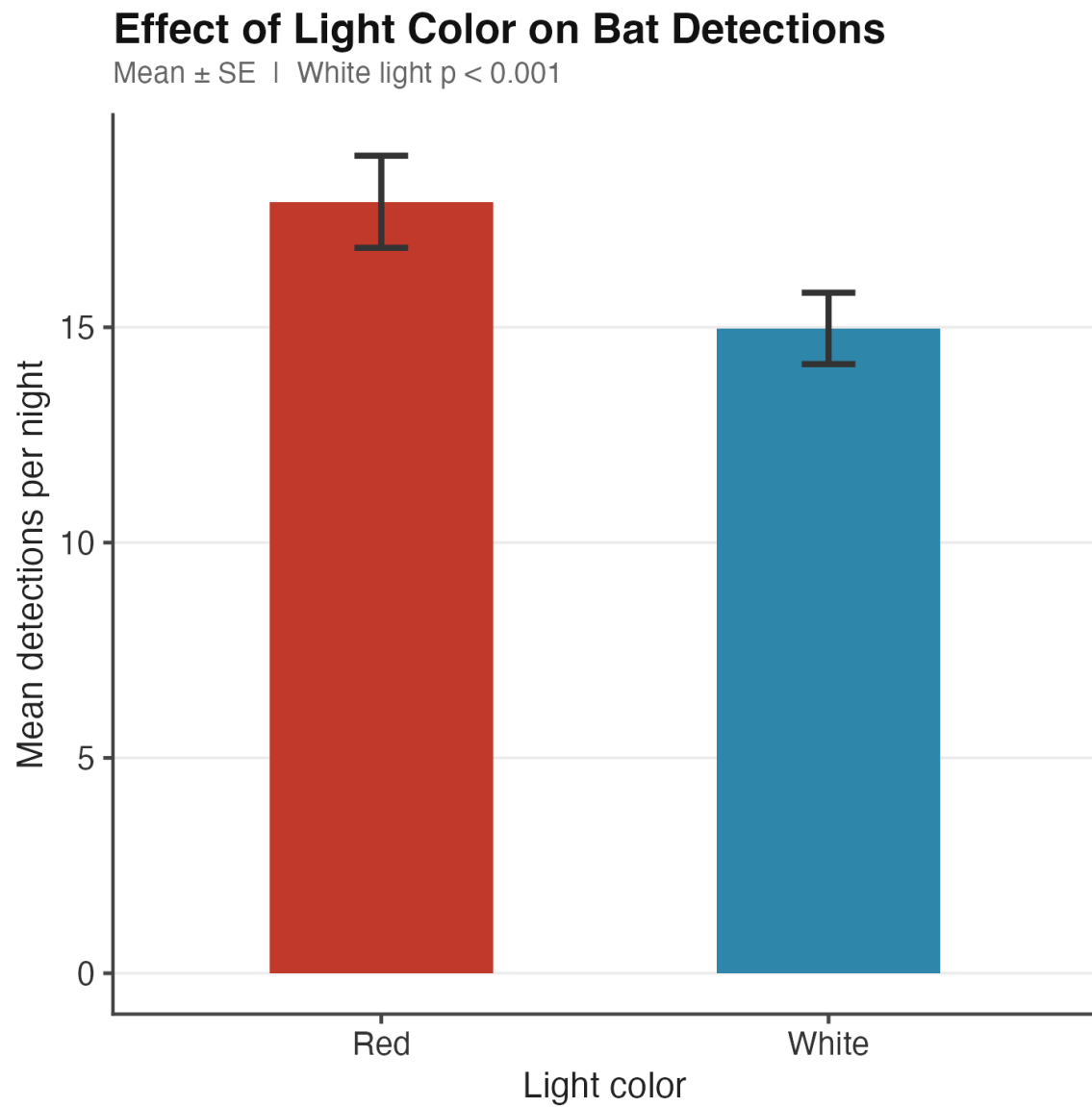


Figure 1: Mean detections per night by light colour, averaged across intensities. Error bars = SE.

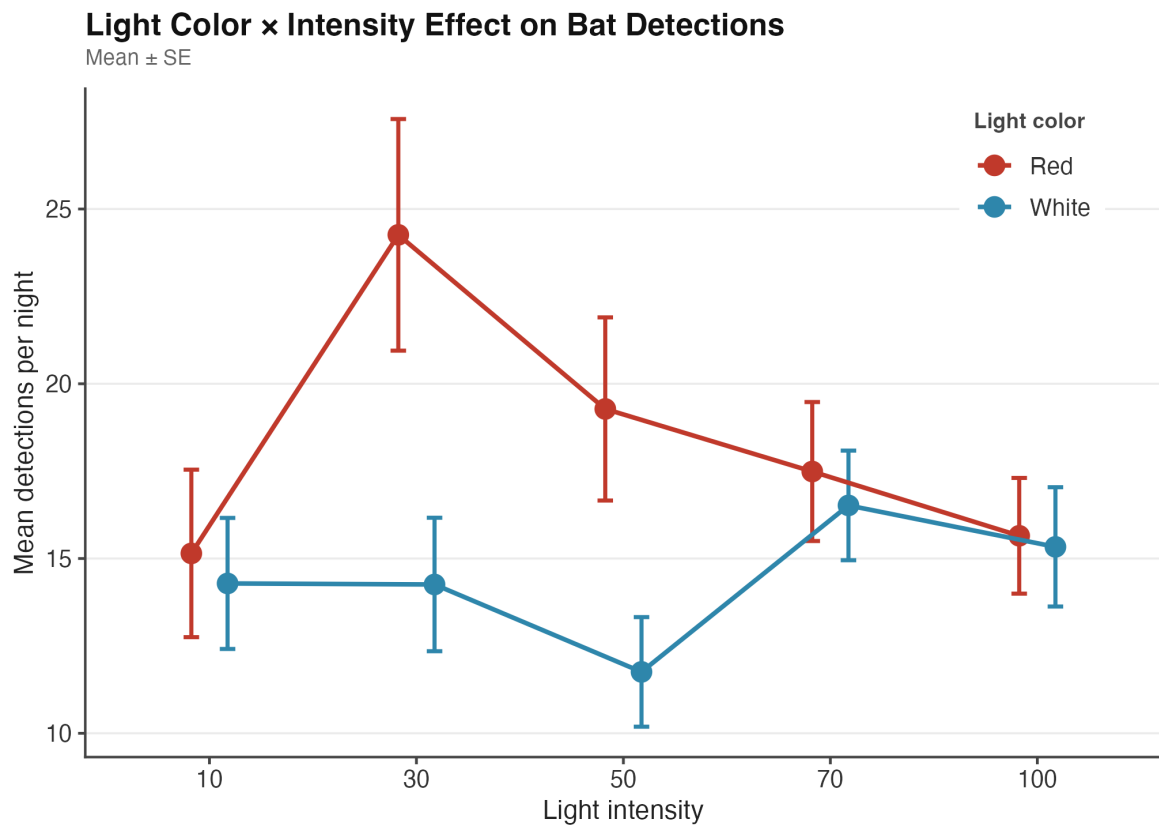


Figure 2: Mean detections per night by colour at each intensity level with 95% CI. Red peaks at 30% and converges with white at high intensity.

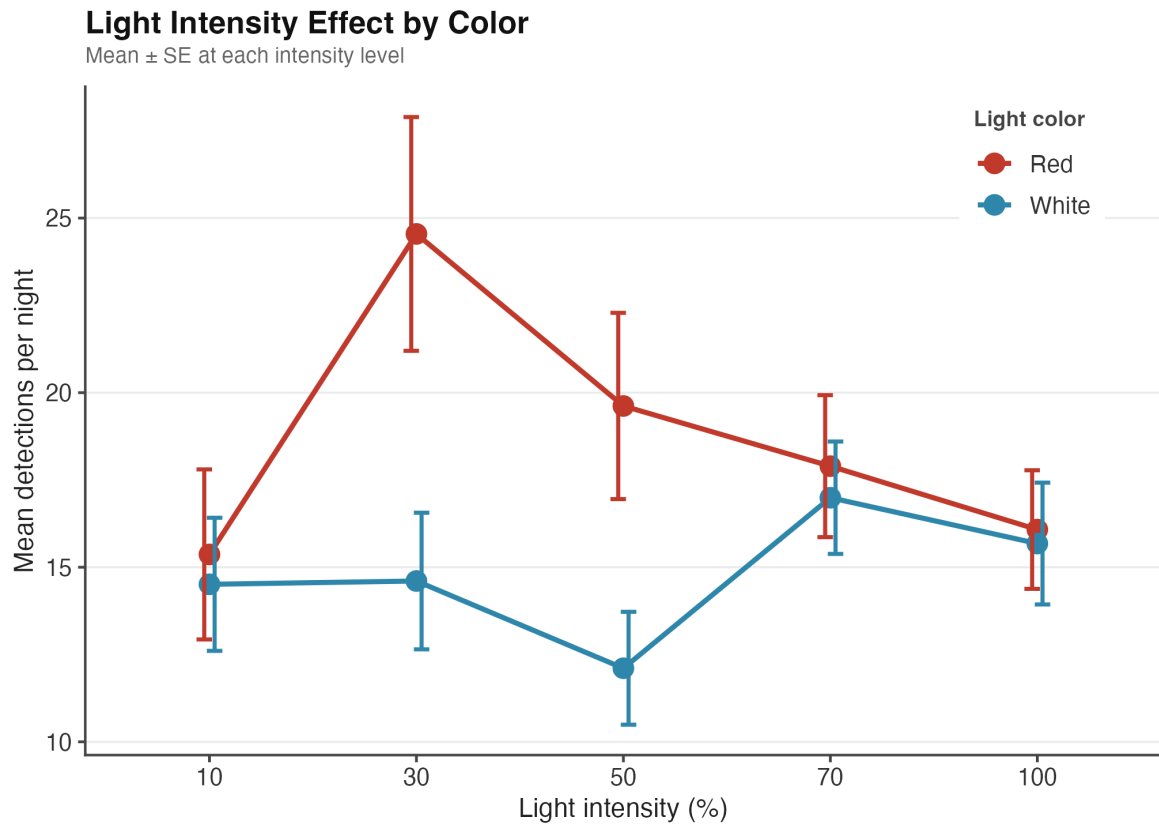


Figure 3: Colour $\times$ intensity interaction: predicted detections from the GLMM on the response scale.

POST-HOC: INTENSITY WITHIN COLOUR

=== Intensity pairwise contrasts within Red (Holm-corrected) ===

contrast	ratio	SE	df	z.ratio	p.value	
30 / 10	1.815	0.248	Inf	4.365	<.0001	***
50 / 10	1.427	0.204	Inf	2.480	0.0590	
70 / 10	1.425	0.133	Inf	3.794	0.0011	**
100 / 10	0.999	0.116	Inf	-0.002	1.0000	
50 / 30	0.786	0.107	Inf	-1.788	0.3700	
70 / 30	0.785	0.107	Inf	-1.770	0.3700	
100 / 30	0.550	0.076	Inf	-4.362	<.0001	***
70 / 50	0.999	0.096	Inf	-0.012	1.0000	
100 / 50	0.700	0.101	Inf	-2.480	0.0590	
100 / 70	0.701	0.081	Inf	-3.026	0.0148	*

=== Intensity pairwise contrasts within White (Holm-corrected) ===

contrast	ratio	SE	df	z.ratio	p.value
30 / 10	0.880	0.081	Inf	-1.396	1.0000
50 / 10	0.873	0.080	Inf	-1.467	1.0000
70 / 10	1.260	0.117	Inf	2.495	0.1266
100 / 10	1.097	0.101	Inf	1.006	1.0000

[all remaining contrasts: p >= 0.18]

Red peaks at 30% (1.82x above 10%; $p < 0.0001$) and collapses at 100% (0.55x relative to 30%; $p < 0.0001$). White is flat across all intensities (no pairwise contrast survives Holm correction).

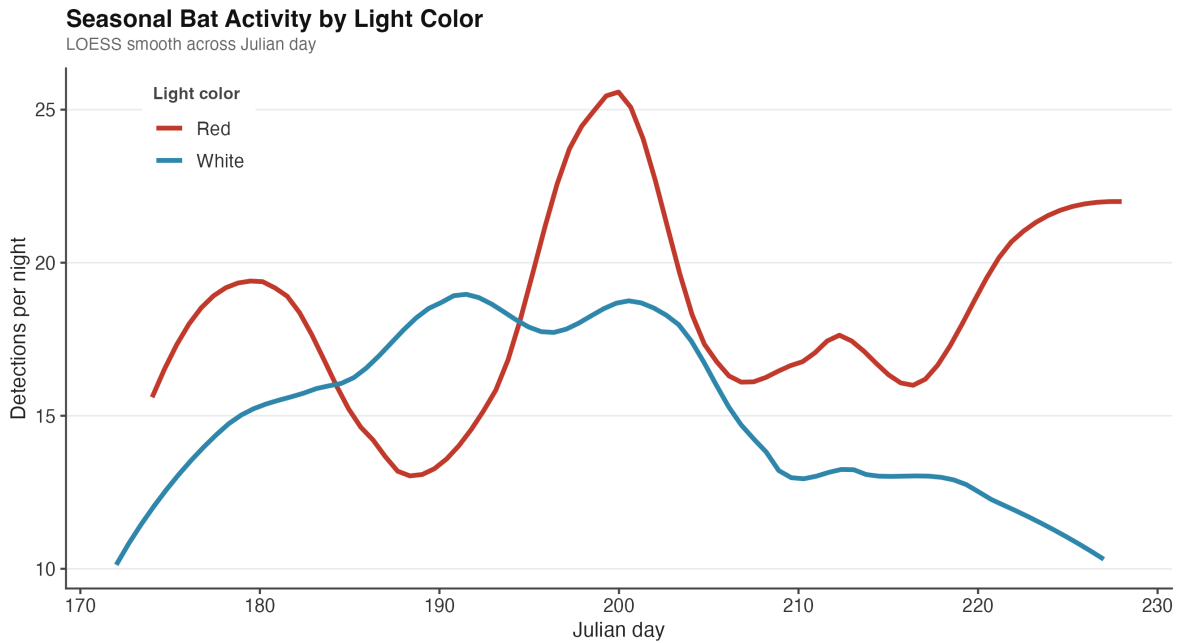


Figure 4: Seasonal trend in bat detections by light colour. LOESS smoothed; shading = 95% CI. Both colours decline through the season; red/white separation persists across survey period.

Effect of Moon Phase on Bat Detections

LOESS smooth by light color (p = 0.008)

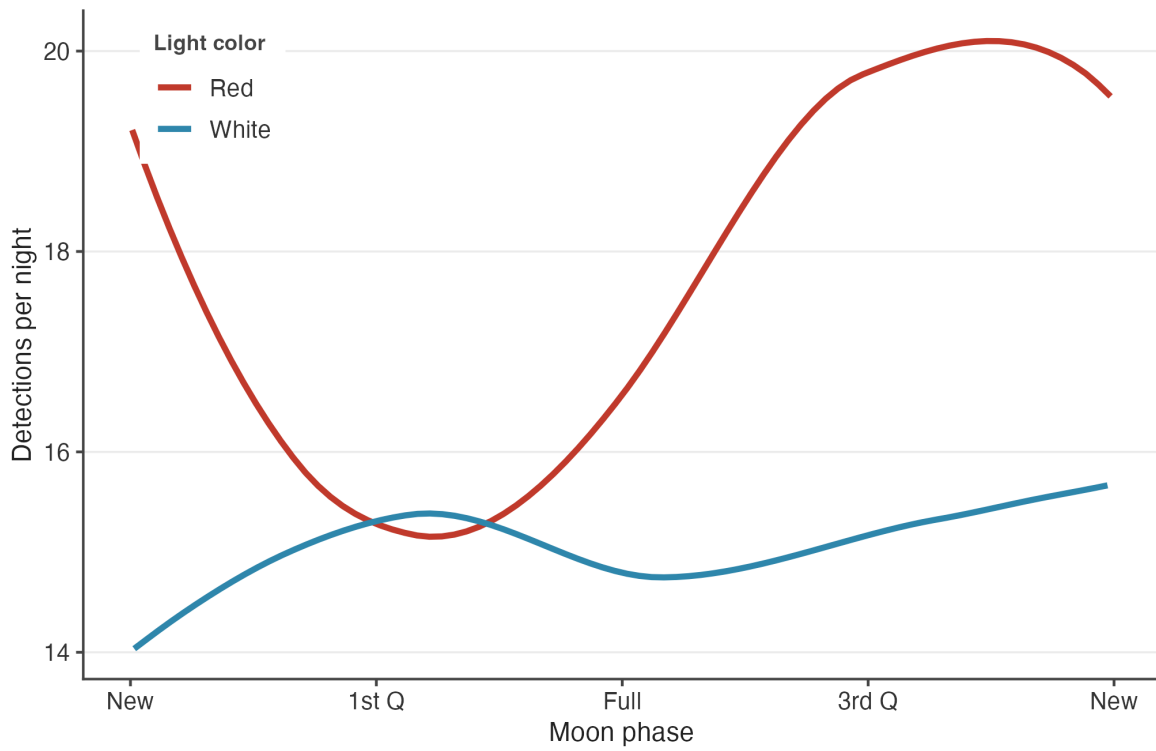


Figure 5: Bat detections vs. moon phase by colour. No evident colour $\times$ moon interaction.

POST-HOC: DISTANCE

=== Estimated marginal means: distance categories ===

dist_cat	response	SE	df	asympt.LCL	asympt.UCL
Close	6.5	2.8	Inf	2.9	14.2
Medium	21.4	9.9	Inf	9.0	50.5
Further	21.6	7.0	Inf	11.7	40.0
Far	17.4	6.2	Inf	8.8	34.2

=== Distance pairwise contrasts (Holm-corrected) ===

contrast	ratio	SE	df	z.ratio	p.value
Medium / Close	3.296	2.097	Inf	1.876	0.4833
Further / Close	3.336	1.725	Inf	2.330	0.2967

Far / Close	2.686	1.503	Inf	1.761	0.4833
Further / Medium	1.012	0.590	Inf	0.020	1.0000
Far / Medium	0.815	0.462	Inf	-0.364	1.0000
Far / Further	0.806	0.424	Inf	-0.415	1.0000

Close sites had 3x fewer detections than all other categories. Medium, Further, and Far were indistinguishable. No pairwise contrast survives Holm correction (low power with $n=14$ sites), but the main model confirms the distance effect (dist_catFar: $p = 0.040$, dist_catFurther: $p = 0.044$).

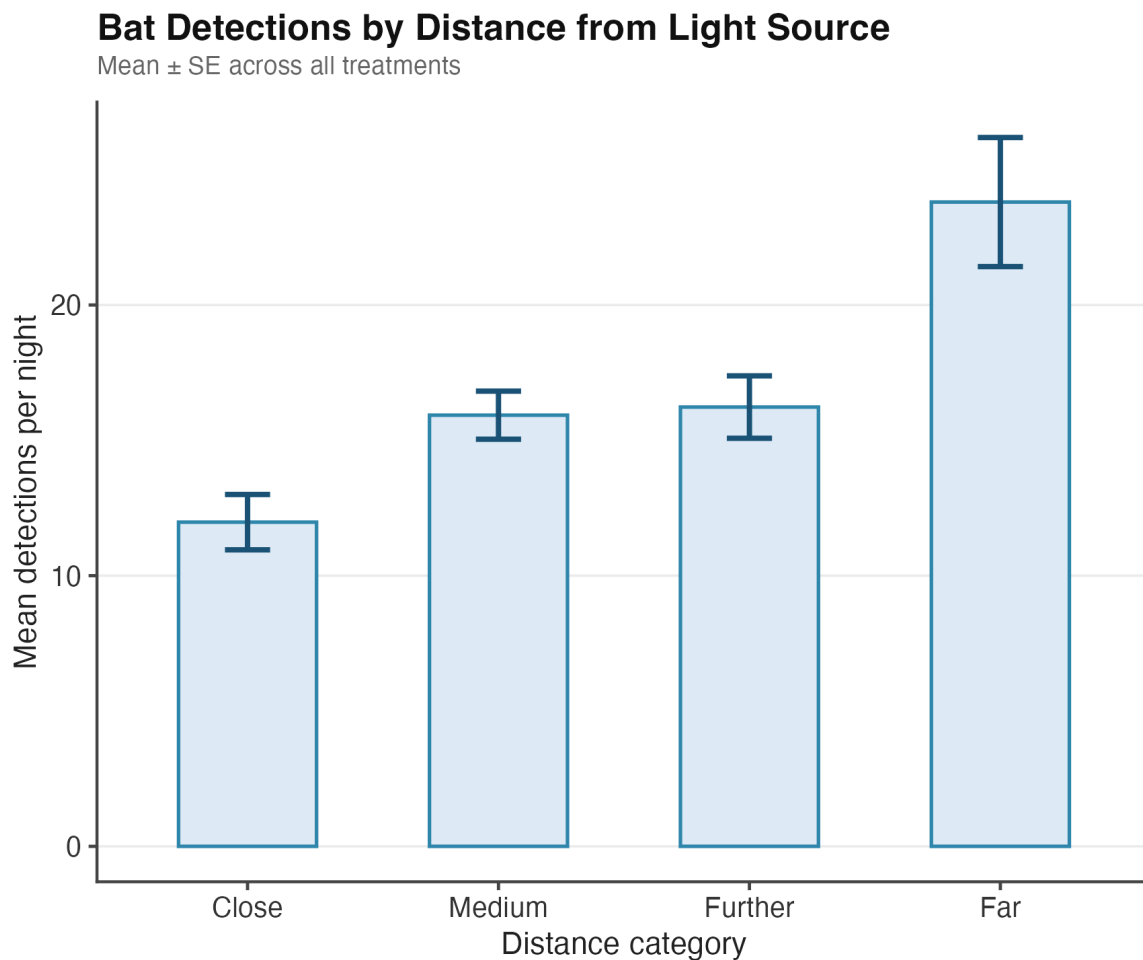


Figure 6: Mean detections per night by distance category. Error bars = SE.

Bat Detections vs Distance from Light Source

Site-level means with linear trend (Spearman correlation)

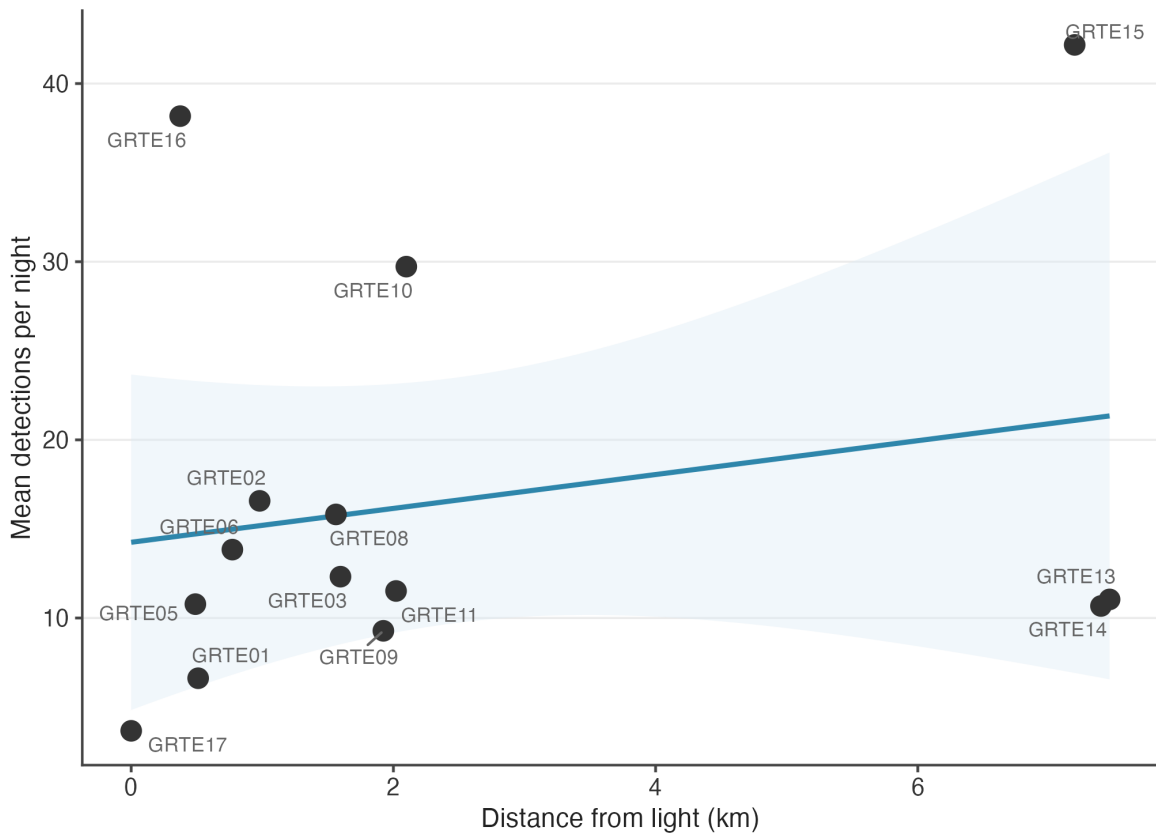


Figure 7: Site-level mean detections vs. distance from light source (km) with linear trend and Spearman correlation.

Detections by Distance, Color, and Intensity

Mean $\pm$ SE

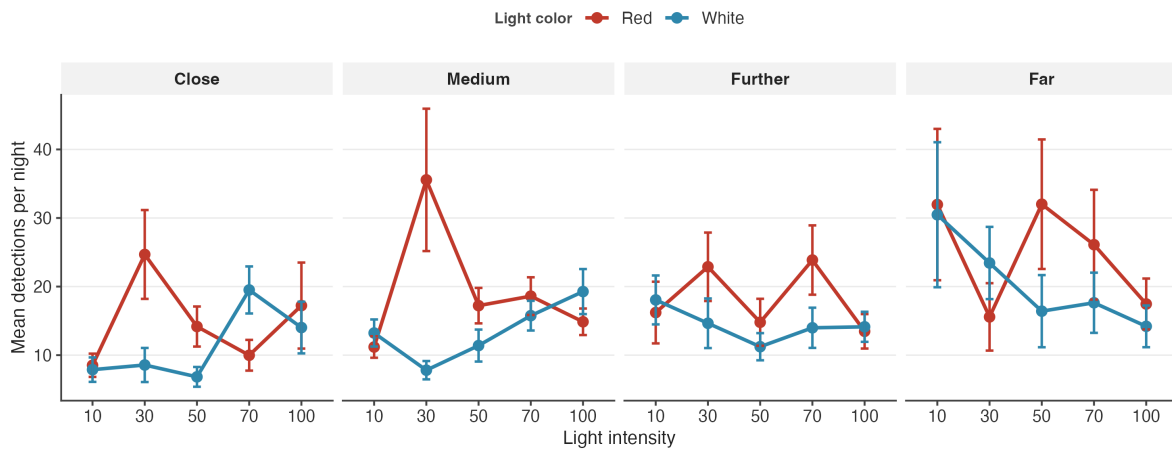


Figure 8: Detections by distance category and light treatment.

COMMUNITY ANALYSIS

=== NMDS ===

Stress: 0.207 (k = 2, Bray-Curtis, trymax = 100)

=== PERMANOVA (adonis2) ===

	Df	SumOfSqs	R2	F	Pr(>F)	
color	1	0.357	0.013	0.657	0.196	
intensity	1	0.920	0.033	1.693	0.046	*
dist_cat	3	4.366	0.157	2.678	<0.001	***
mean_phase	1	0.468	0.017	0.861	0.488	
pct_nonforest	1	4.087	0.147	7.527	<0.001	***
brightness_d	1	2.281	0.082	4.199	0.003	**
Residual	38	20.648	0.744			
Total	46	27.807	1.000			

R-squared (total model): 0.256

=== Homogeneity of dispersion (by color) ===

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Groups	1	0.0007	0.00071	0.00629	0.938
Residuals	45	5.0815	0.11292		

=== dbRDA marginal tests ===

	F	Pr(>F)	
color	1.45	0.196	
intensity	1.69	0.046	*
dist_cat	8.03	<0.001	***
mean_phase	0.86	0.488	
pct_nonforest	7.53	<0.001	***
brightness_d	4.20	0.003	**

=== mvabund -- multivariate GLM (nBoot = 9999) ===

	Dev	Df	p.uni (adjusted)	
color	9.2	1	0.394	
intensity	83.5	4	0.002	**
dist_cat	311.9	3	<0.001	***
mean_phase	13.2	1	0.197	
pct_nonforest	42.2	1	<0.001	***

brightness_d 57.4 1 <0.001 ***

Colour does not restructure community composition. Distance does (R2 contribution largest single predictor). Dispersion homogeneous between colour treatments (p = 0.938) -- PERMANOVA result reflects location, not spread.

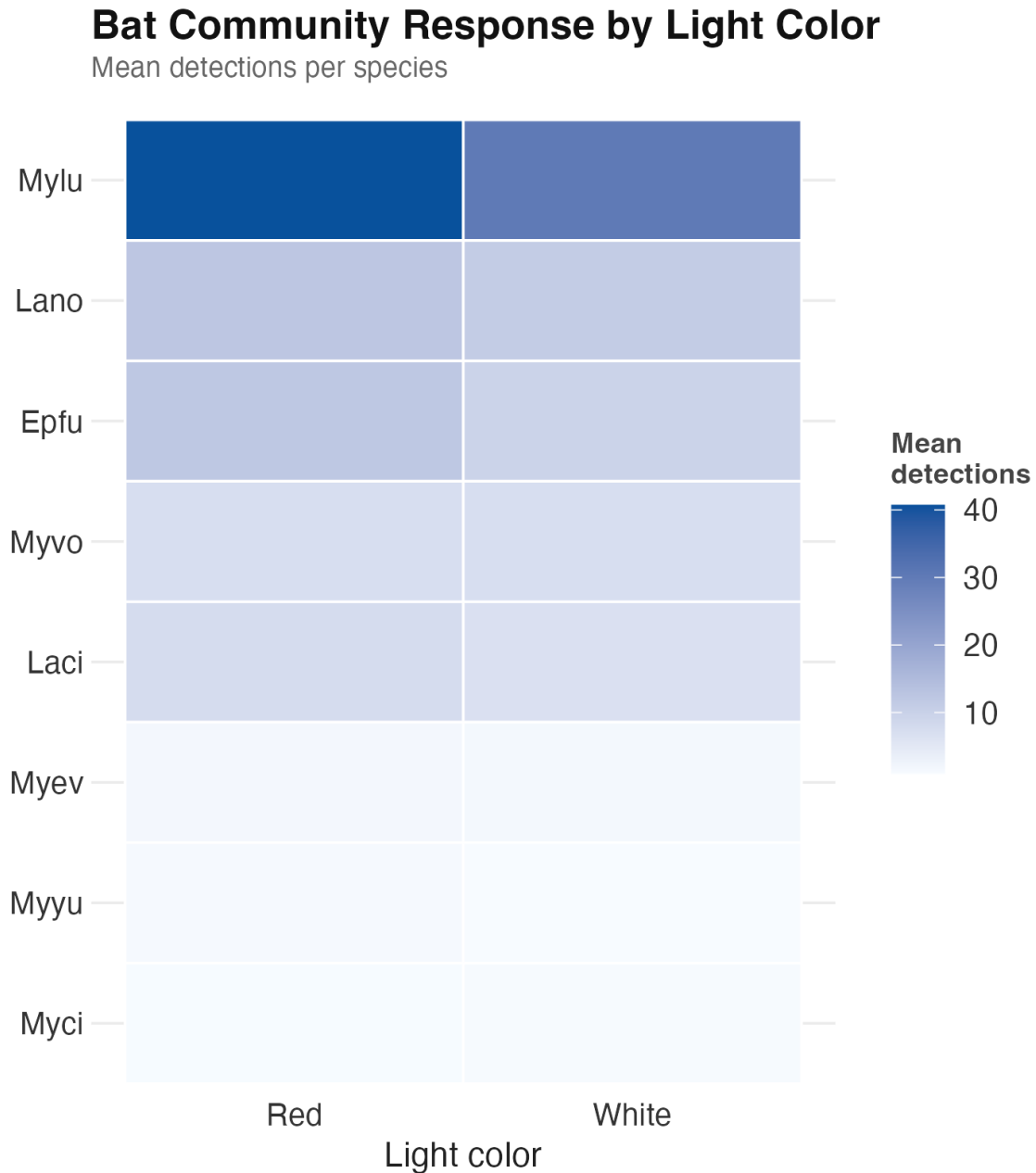


Figure 9: Species activity heatmap (z-scored within species) by light colour and intensity.

SPECIES-LEVEL RESULTS

=== mvabund univariate p-values (adjusted) ===

Species	Colour	Intensity	Distance	%nonforest	brightness
Epfu	ns	***	***	ns	ns
Laci	ns	.	**	ns	ns
Lano	ns	ns	**	ns	ns
Myev	ns	ns	***	.	ns
Mylu	ns	ns	**	***	ns
Myvo	ns	ns	***	ns	***
Myyu	ns	ns	ns	ns	ns
Myci	ns	ns	ns	ns	ns

Epfu: most treatment-sensitive; responds to both intensity and distance.
Mylu: tracks habitat (pct_nonforest). Myvo: tracks sky darkness. Myyu,
Myci: no significant response to any measured predictor.

Species Detections by Distance from Light Source

Mean detections per species × distance category

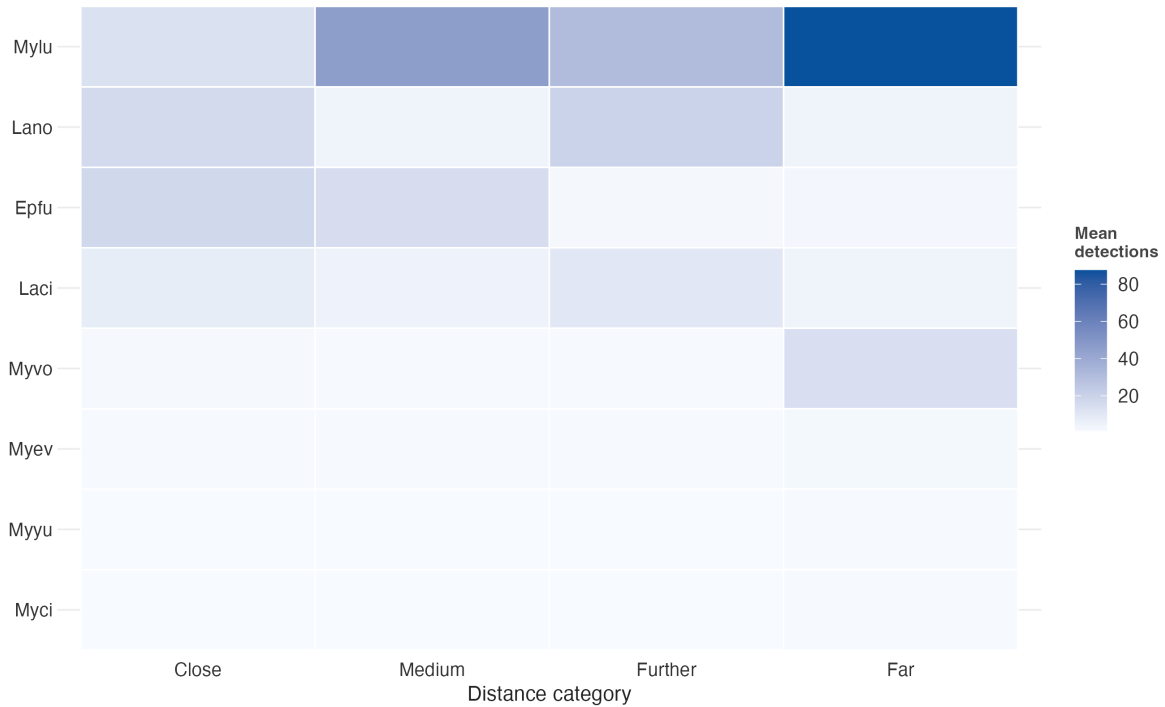
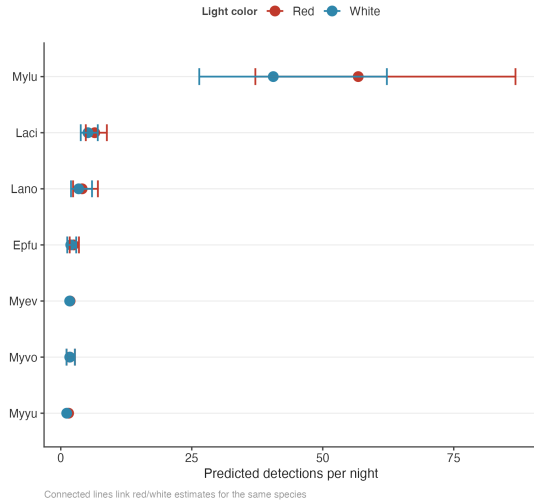


Figure 10: Species activity by distance category (z-scored within species).

Species-Level Light Effects — Distance Project

Light Color Effect on Bat Detections by Species
Marginal means $\pm$ 95% CI from per-species negative binomial GLMMs



White vs. Red Light — Detection Ratio by Species & Intensity
Ratio > 1 = more detections under white; < 1 = more under red

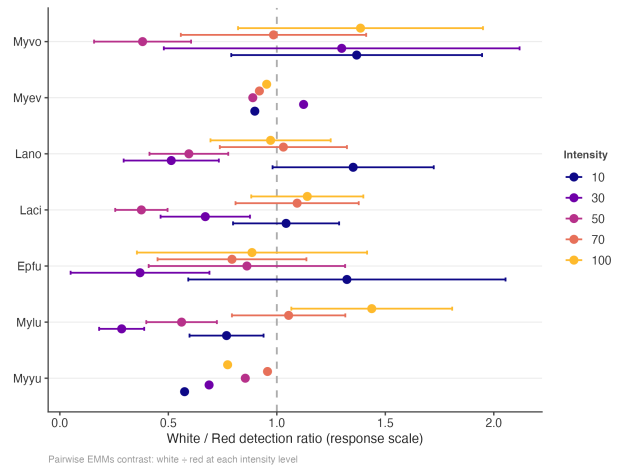


Figure 11: Left: marginal means $\pm$ 95% CI by colour, ranked by species. Right: white/red detection ratio by intensity level per species. Ratio > 1 = more detections under white; < 1 = more under red.

White vs. Red Light Effect by Species and Intensity

Negative values (red) = white light suppresses relative to red

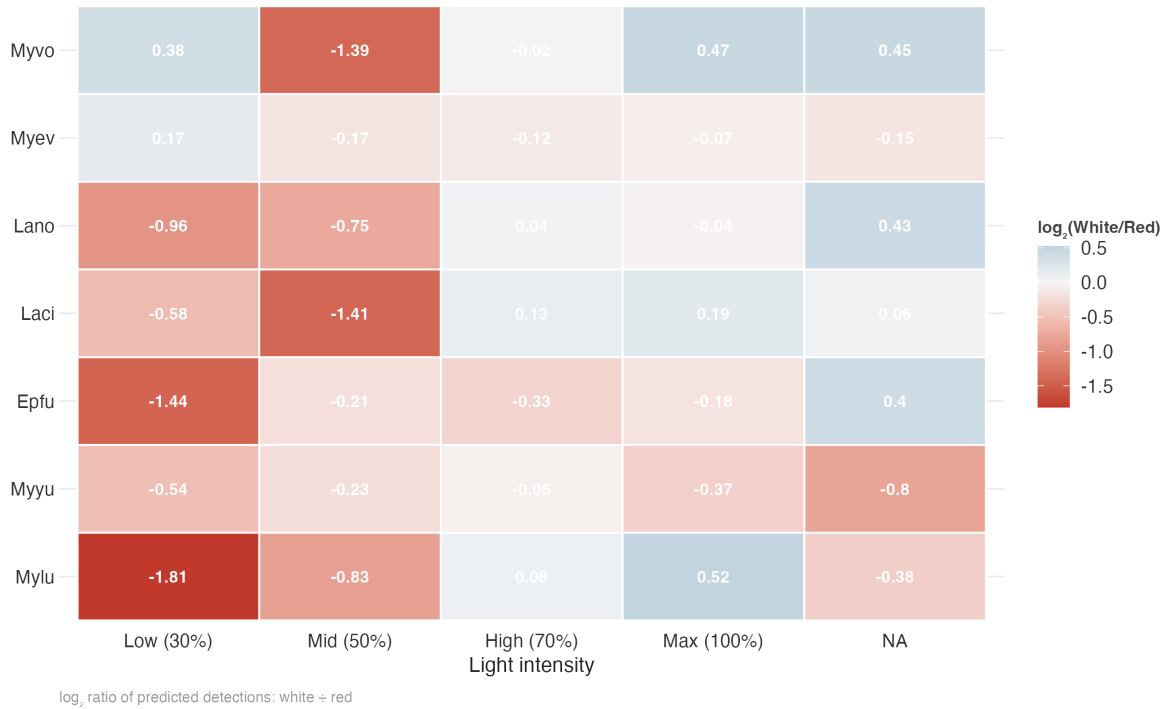


Figure 12: Log₂(white/red) heatmap by species and intensity. Negative values (red) = white suppresses relative to red.

SUMMARY

1. Colour x intensity interaction: red light at 30-50% produced 1.6-2.1x more detections than white. At 70% and 100%, colours converge. White light is flat at all intensities.
2. Distance: sites within 0.5 km had 3x fewer detections. Recovery appears complete by 0.5-1.0 km.
3. Community composition is driven by distance, habitat, and sky darkness -- not by light colour.
4. Management implication: red light at 30-50% intensity, sited >0.5 km from sensitive habitat, maximises bat detections. White light suppresses activity at any intensity.

SPECIES KEY

Epfu	<i>Eptesicus fuscus</i>	(big brown bat)
Laci	<i>Lasiurus cinereus</i>	(hoary bat)
Lano	<i>Lasionycteris noctivagans</i>	(silver-haired bat)
Myci	<i>Myotis ciliolabrum</i>	(western small-footed myotis)
Myev	<i>Myotis evotis</i>	(long-eared myotis)
Mylu	<i>Myotis lucifugus</i>	(little brown myotis)
Myvo	<i>Myotis volans</i>	(long-legged myotis)
Myyu	<i>Myotis yumanensis</i>	(Yuma myotis)